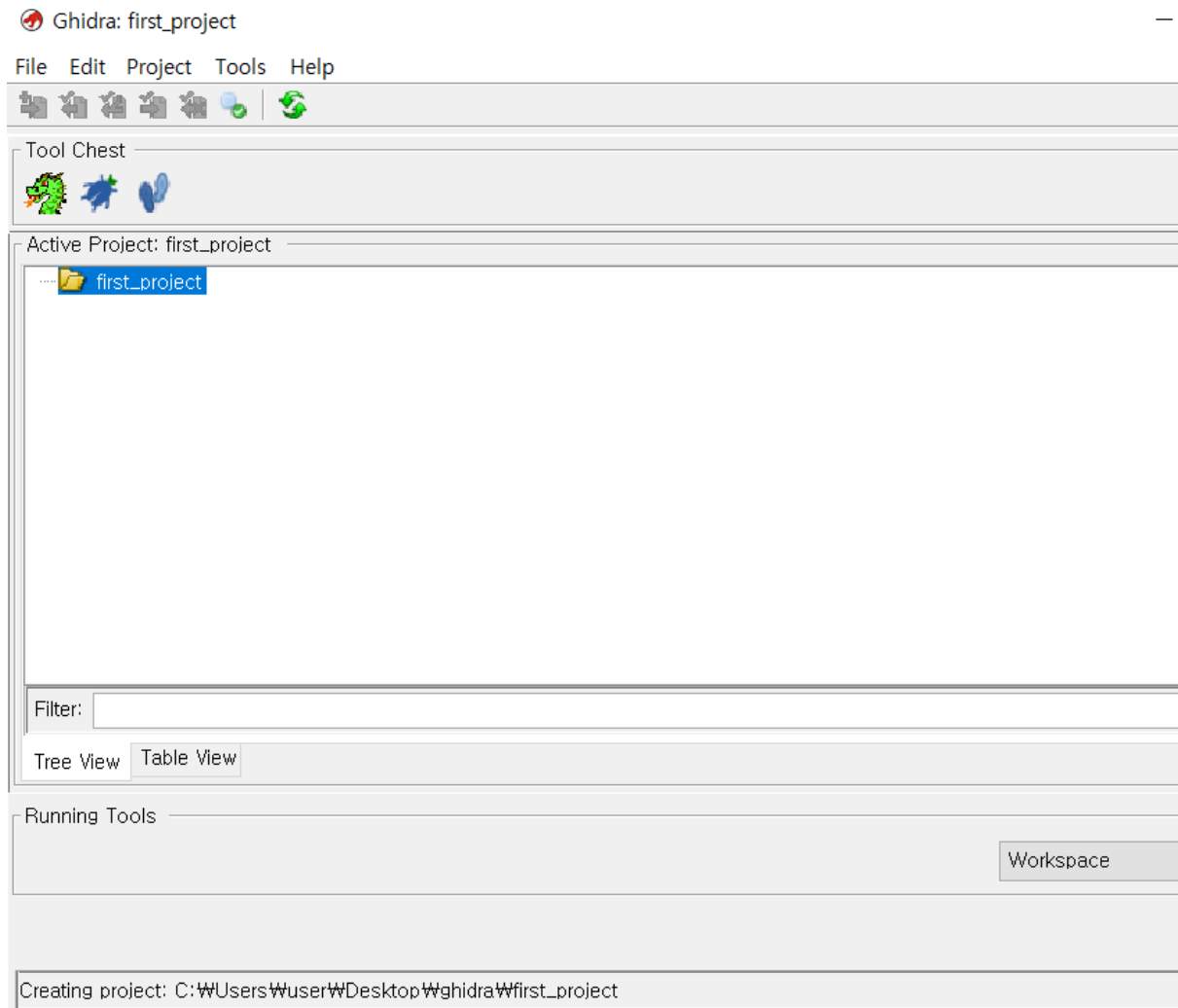




코드 브라우저



- 코드 브라우저 (Dragon)

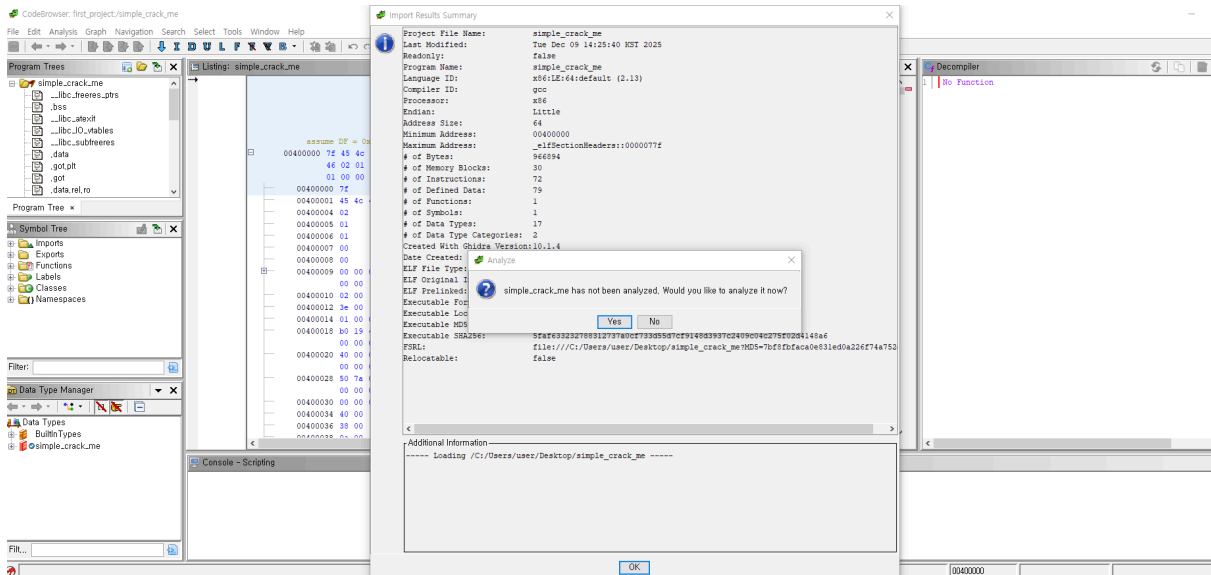
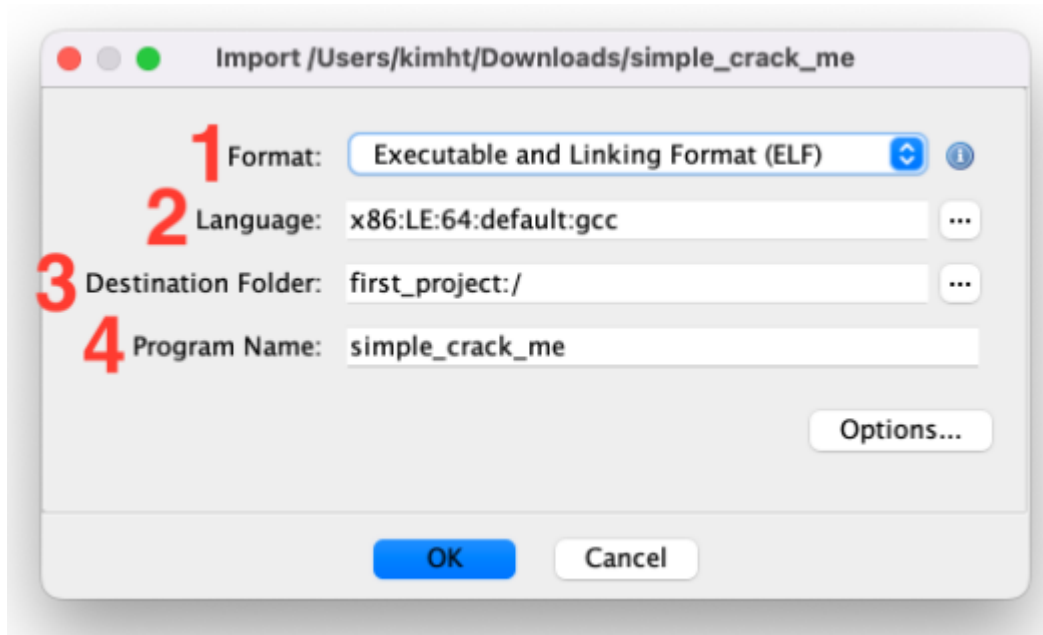
- 디어셈블
- 디컴파일

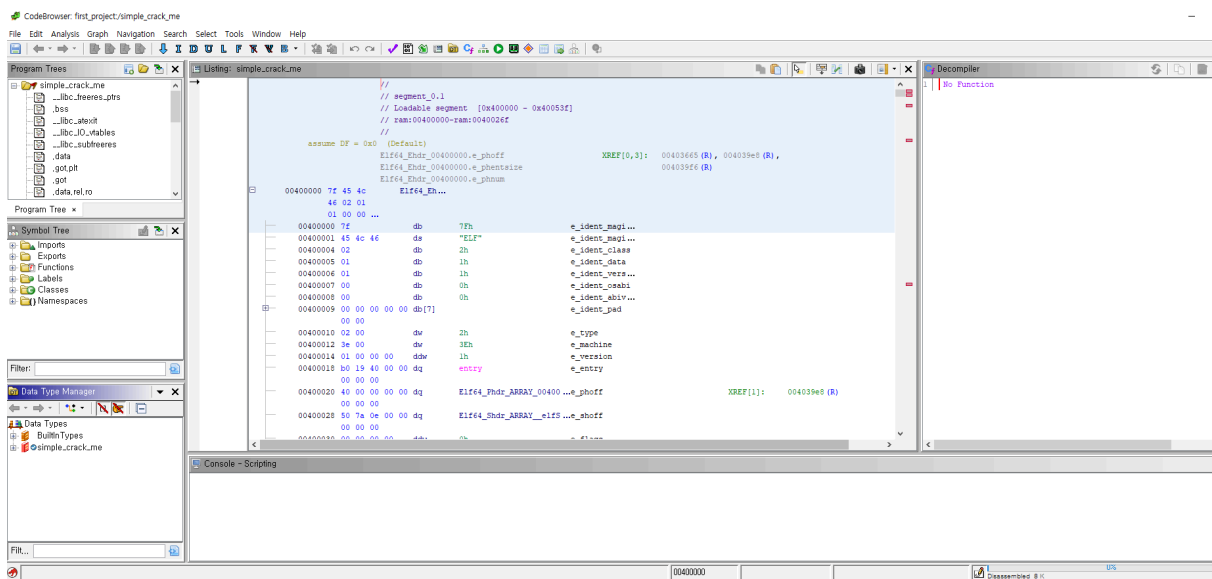
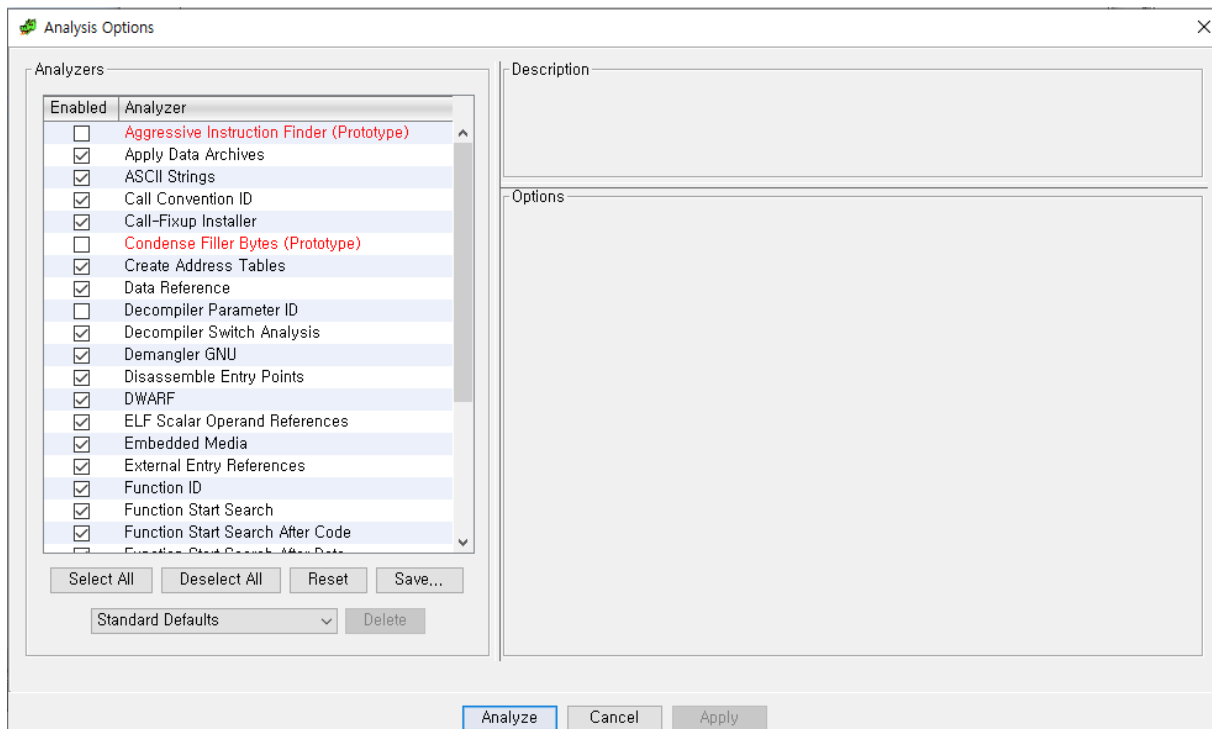
- 디버거 (Bug)

- 디버그

```
root@23NB0068:/mnt/c/Users/user/Desktop# gcc -O0 -no-pie -static -s simple_crack_me.c -o simple_crack_me
root@23NB0068:/mnt/c/Users/user/Desktop# gcc -O0 -no-pie -static -s simple_crack_me.c -o simple_crack_me.c
```

Windows에서 실행 원할 시, WSL Ubuntu로 gcc 실행하여 PE 파일을 ELF 파일로 변환

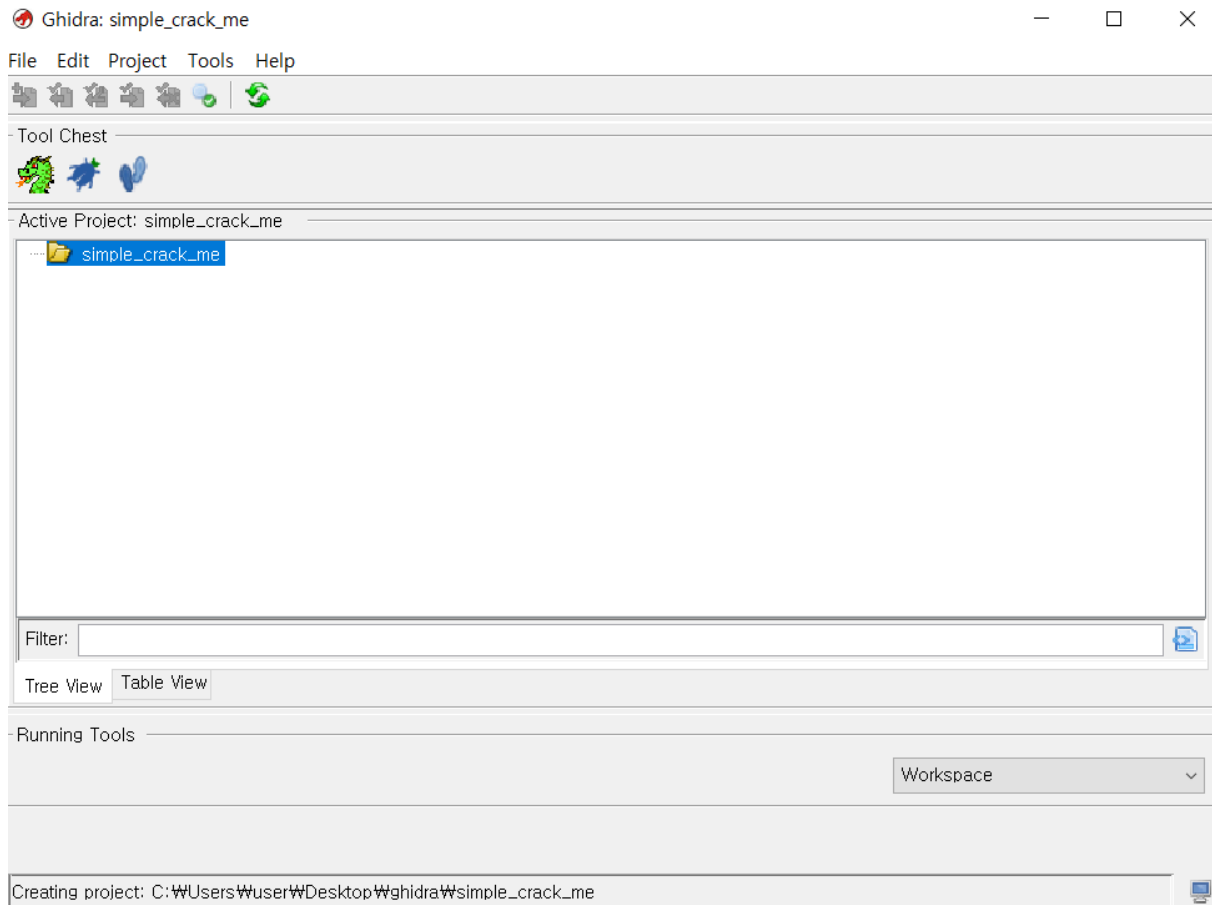




우하단 100% 완료 되면 분석 완료

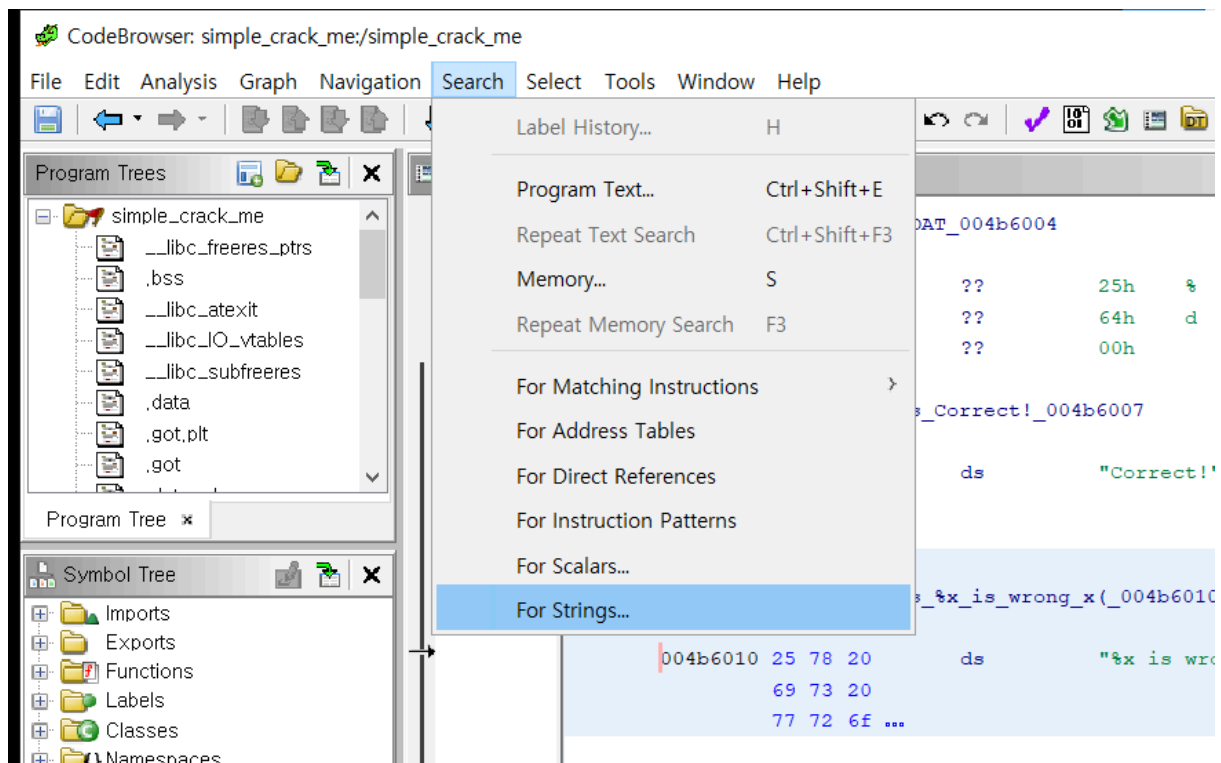


Simple_crack_me

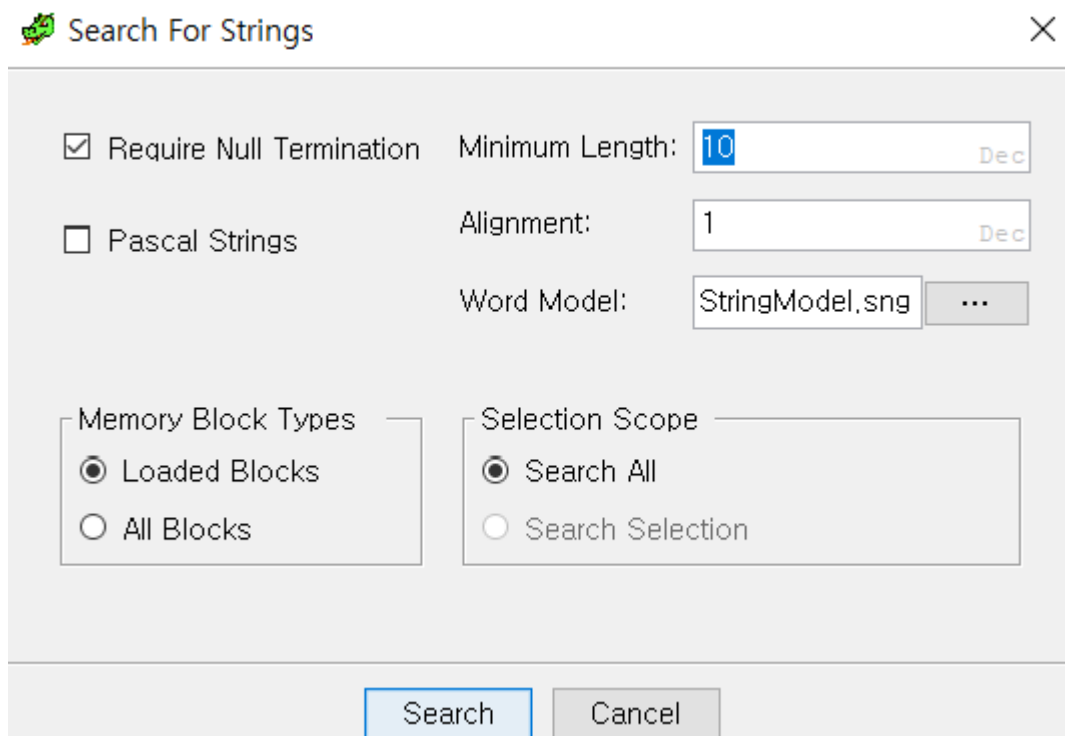


프로젝트 생성

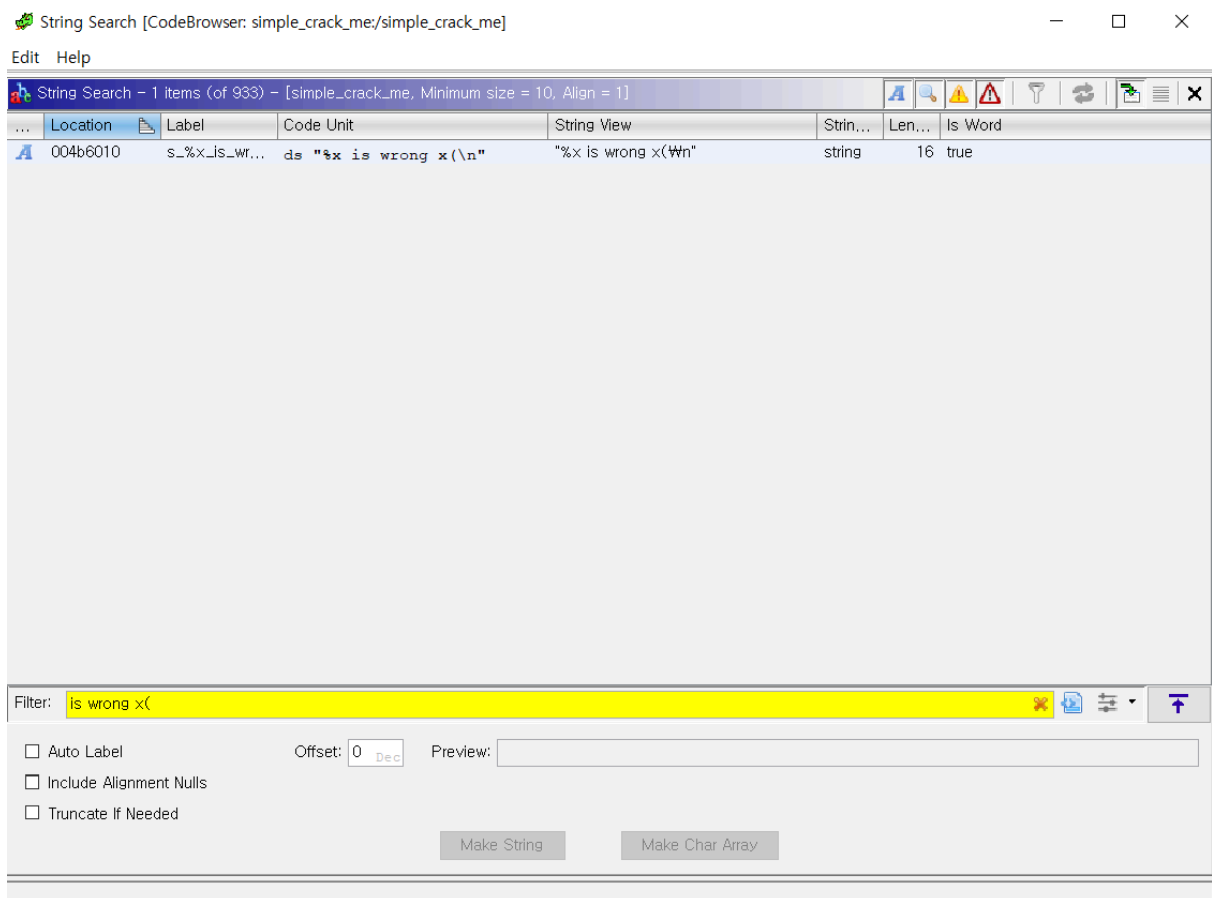
simple_crack_me.c 다시 import



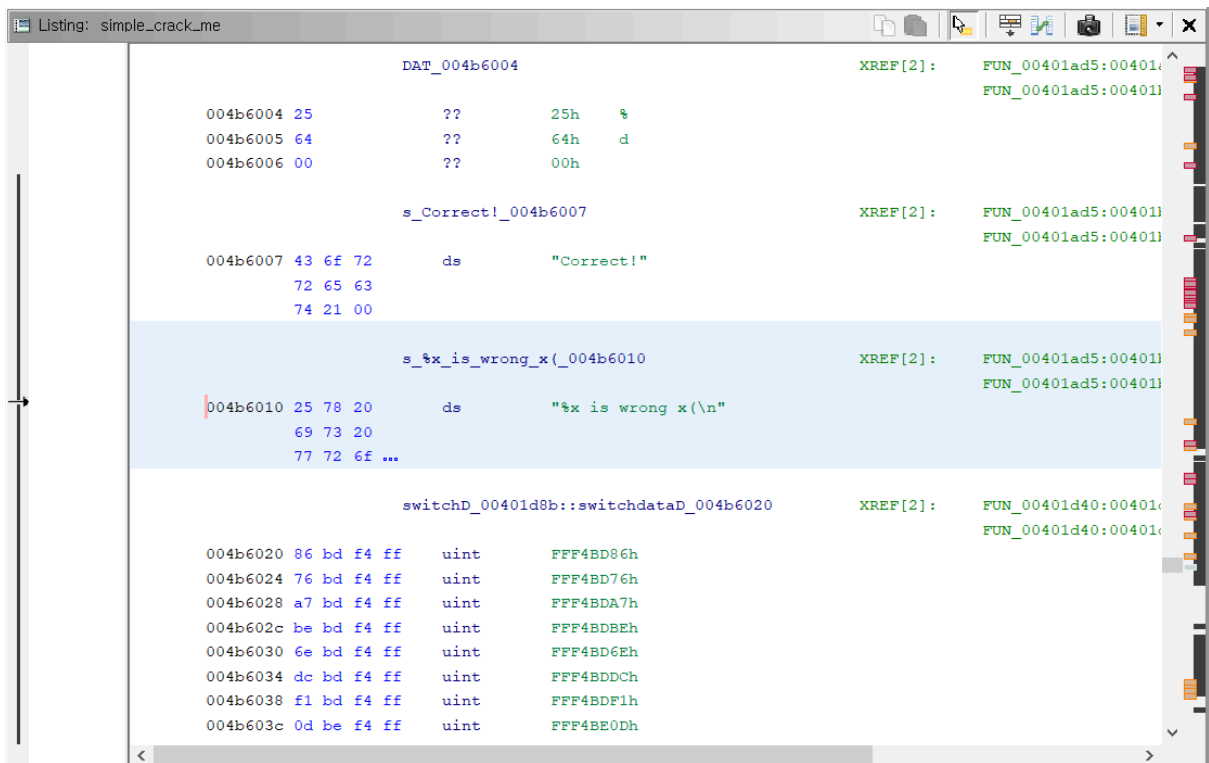
For Strings 클릭



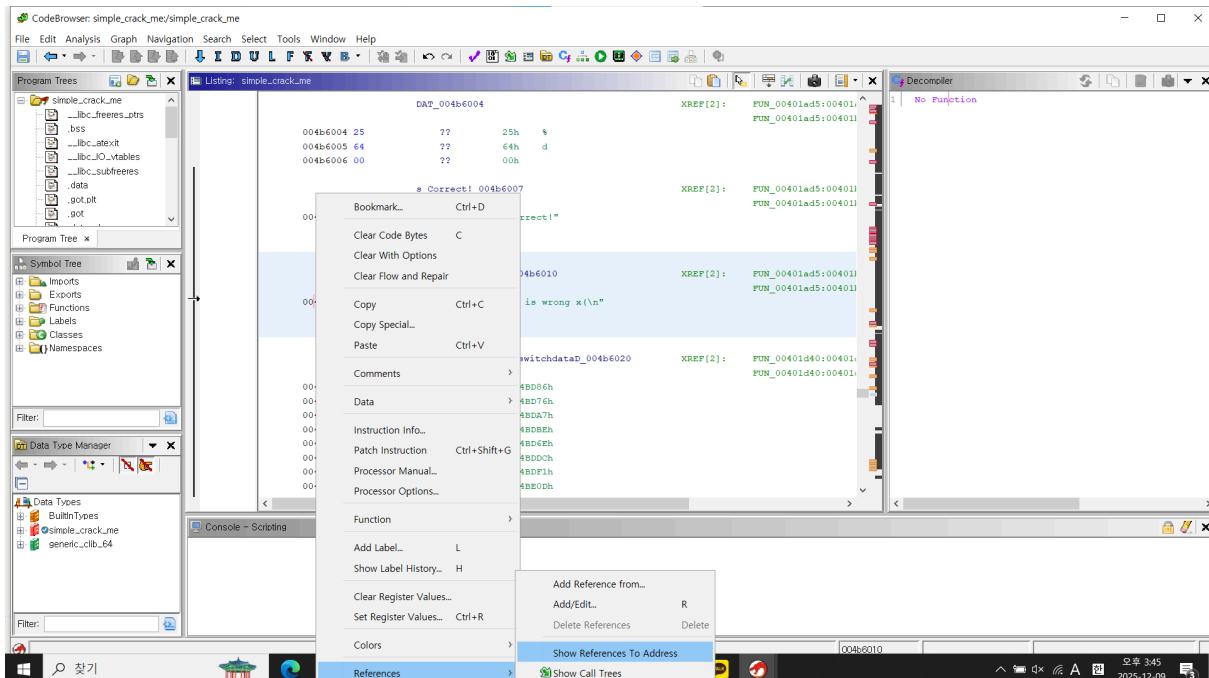
Minimum Length 10으로 변경 후 Search

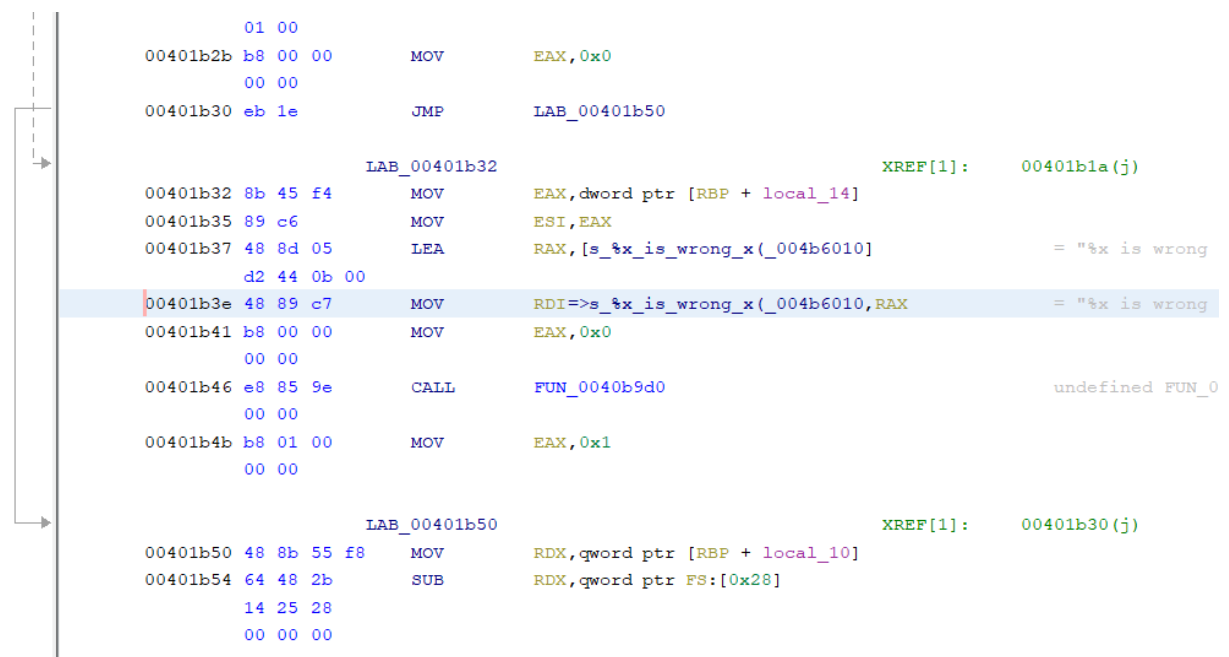
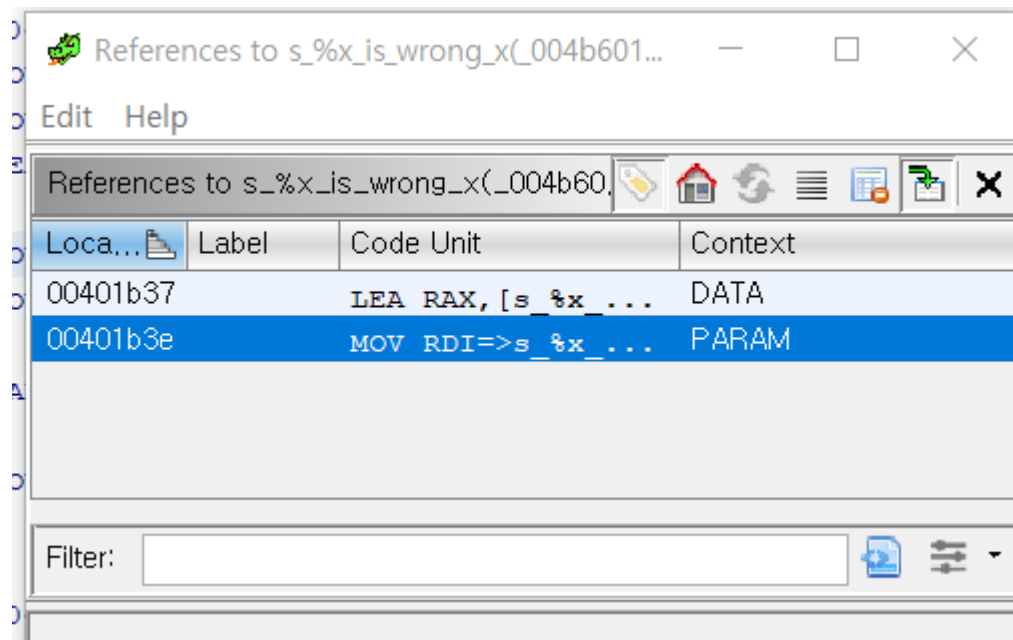


Fliter에 is wrong x(를 입력



위치를 찾아줌





최종적으로 디어셈블 리스팅이 문자열 참조가 발생한 00401b3e 로 이동
 즉 is wrong x(를 출력하는 코드

```
Decompile: FUN_00401ad5 - (simple_crack_me)
1
2 bool FUN_00401ad5(void)
3
4 {
5     long in_FS_OFFSET;
6     bool bVar1;
7     int local_14;
8     long local_10;
9
10    local_10 = *(long *) (in_FS_OFFSET + 0x28);
11    local_14 = 0;
12    FUN_0040bb60(&DAT_004b6004,&local_14);
13    bVar1 = local_14 != 0x13371337;
14    if (bVar1) {
15        FUN_0040b9d0("%x is wrong x(\n",local_14);
16    }
17    else {
18        FUN_0041a440("Correct!");
19    }
20    if (local_10 != *(long *) (in_FS_OFFSET + 0x28)) {
21        /* WARNING: Subroutine does not return */
22        FUN_0045a460();
23    }
24    return bVar1;
25 }
26
```

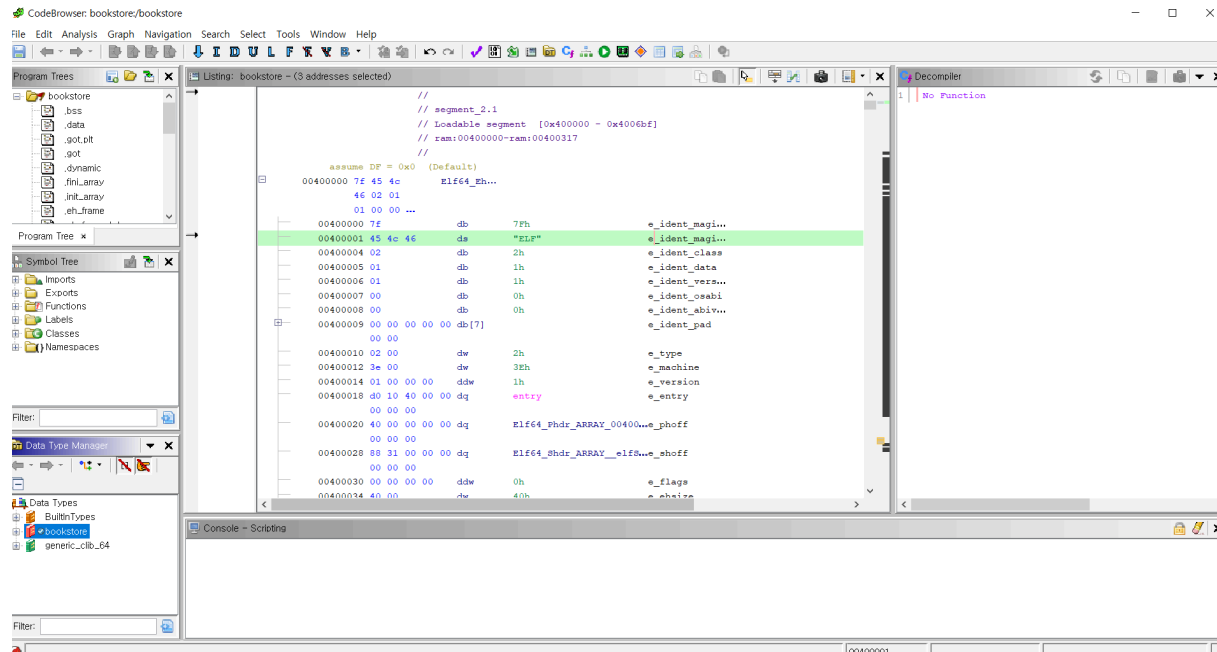
bVar1 값이 0x13371337이 아니라면 `%x is wrong x(` 를 출력

0x13371337이라면 `correct` 출력

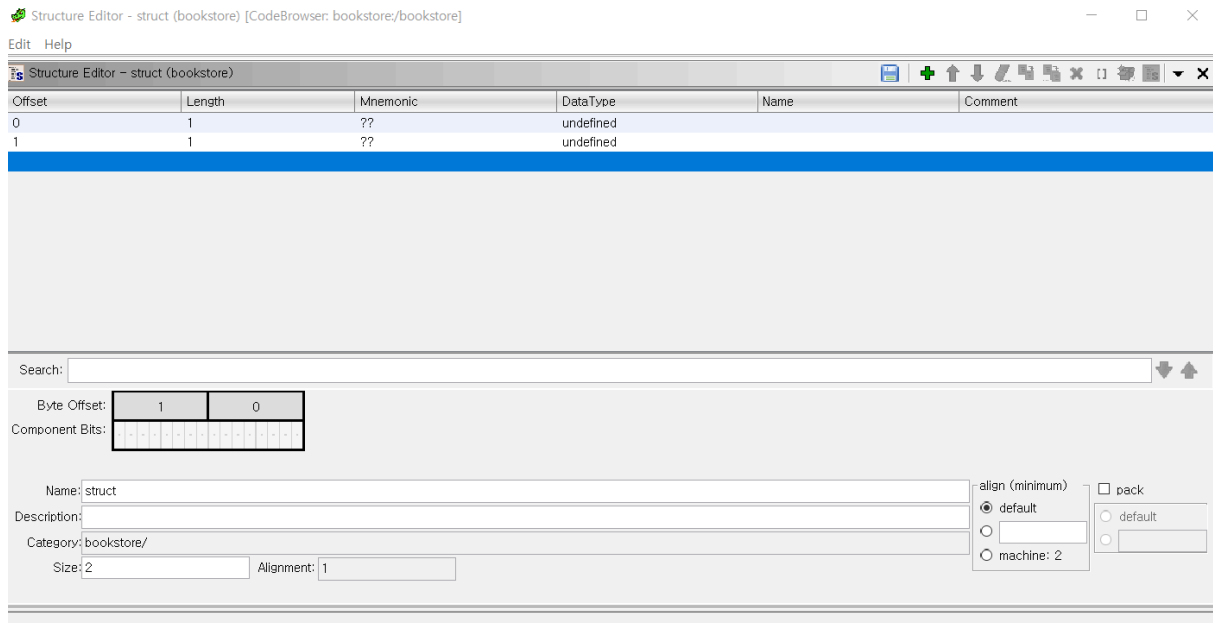
0x13371337를 10진수 변환 시 `322376503` 이므로, `DH{322376503}` 이 정답



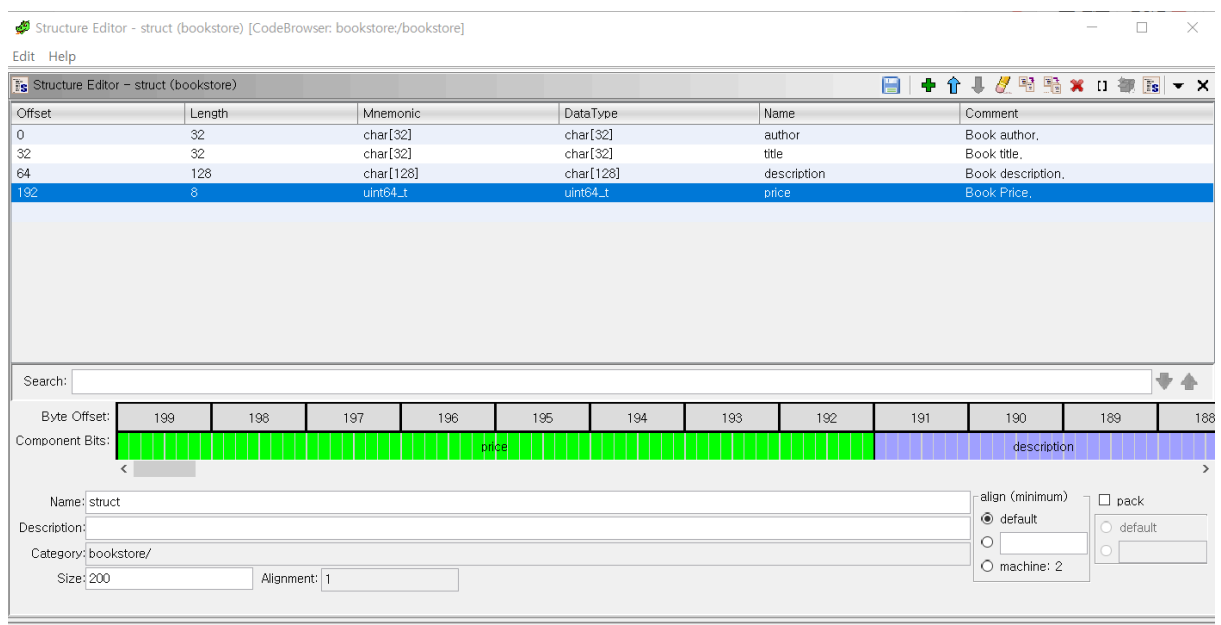
Bookstore



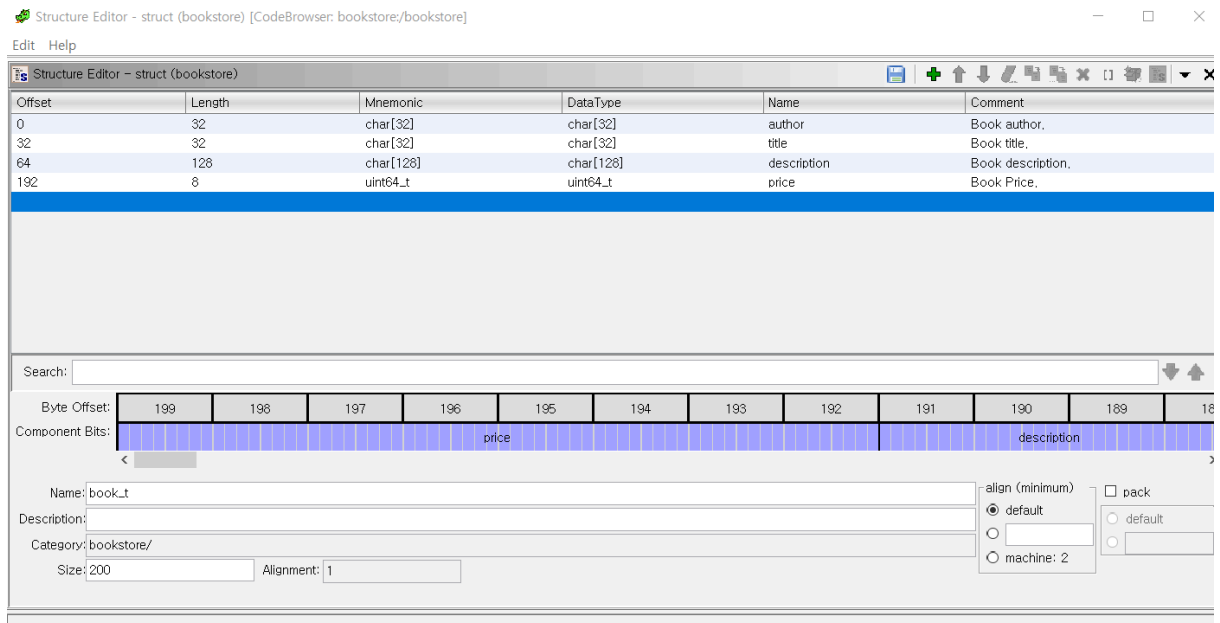
bookstore를 ghidra로 도입한 후, 좌하단의 데이터 타입 매니저에서 bookstore를 오른쪽 클릭한 후 New > Structre 선택하여 새로운 구조체 추가



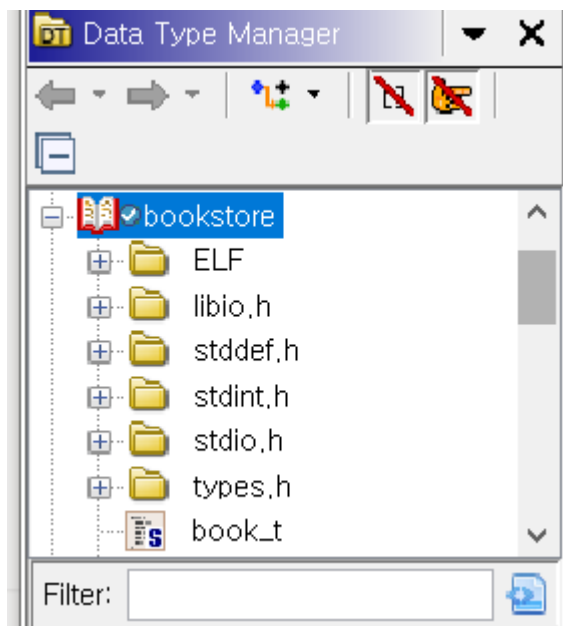
플러스 모양 (추가) 클릭하면 구조체에 필드 추가
디스크 모양 (저장)



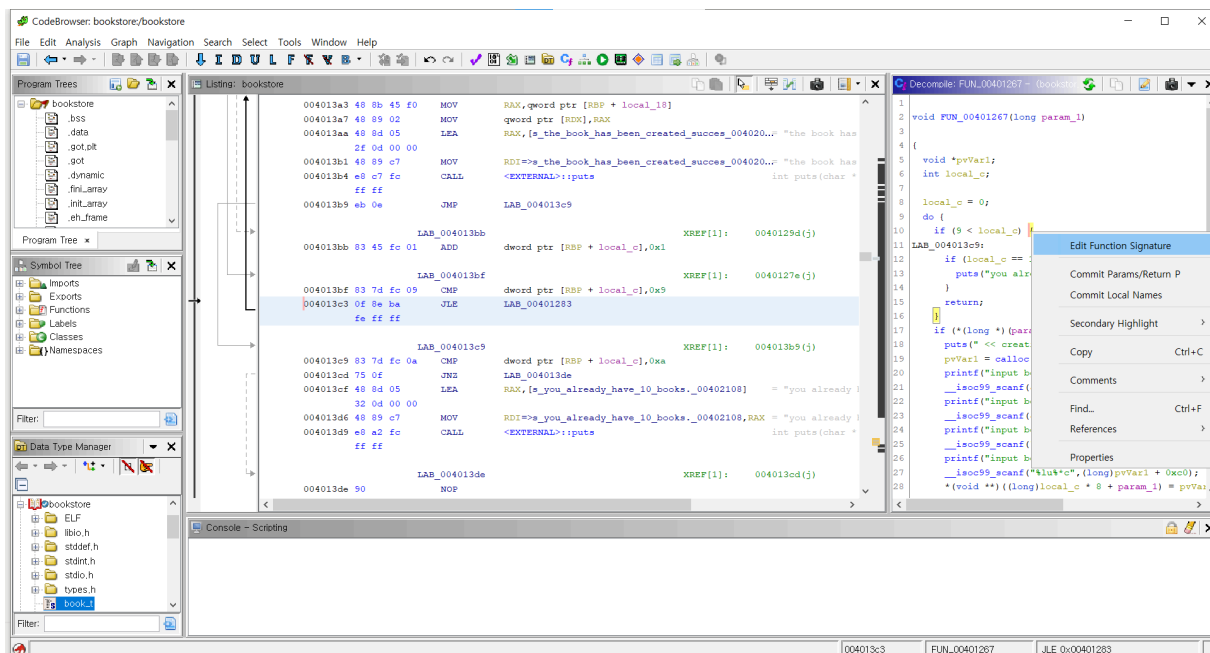
4개의 구조체 설정



name에 `book_t` 설정 후 디스크 이미지 클릭하면 저장



bookstore 마크 다운에 `book_t` 구조체가 생성



Edit Function at 00401267

void `FUN_00401267` (long `param_1`)

Function Name:

`FUN_00401267`

Calling Convention

`__stdcall`

Function Attributes:

☐ Varargs
☐ In Line

☐ No Return
☐ Use Custom Storage

Function Variables

Index	Datatype	Name	Storage
	void	<RETURN>	<VOID>
1	long	<code>param_1</code>	RDI:8

+

×

↑

↓

Call Fixup:

-NONE-

OK

Cancel

```
void CreateBook (book_t * * books)
```

Function Name:

Calling Convention:

Function Attributes:

- ☐ Varargs ☐ In Line
☐ No Return ☐ Use Custom Storage

Function Variables

Index	Datatype	Name	Storage
	void	<RETURN>	<VOID>
1	book_t * *	books	RDI:8



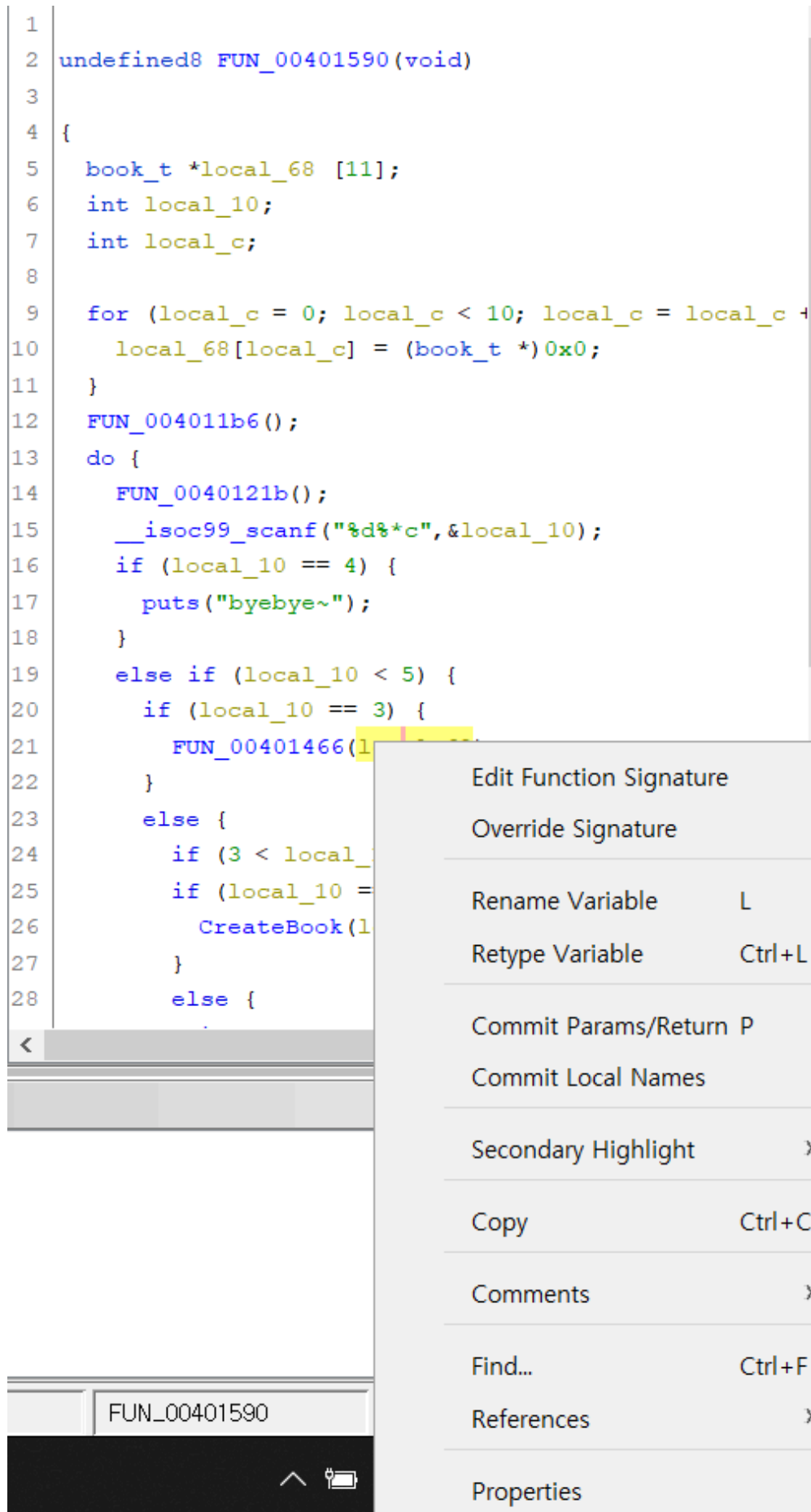
Call Fixup:

OK

Cancel

```
Decompile: CreateBook - (bookstore)

1 void CreateBook(book_t **books)
2
3
4 {
5     book_t *pbVar1;
6     int local_c;
7
8     local_c = 0;
9     do {
10         if (9 < local_c) {
11 LAB_004013c9:
12             if (local_c == 10) {
13                 puts("you already have 10 books.");
14             }
15             return;
16         }
17         if (books[local_c] == (book_t *)0x0) {
18             puts(" << creating book >>");
19             pbVar1 = (book_t *)calloc(1,200);
20             printf("input book title: ");
21             __isoc99_scanf(&DAT_00402086,pbVar1);
22             printf("input book author: ");
23             __isoc99_scanf(&DAT_00402086,pbVar1->title);
24             printf("input book deescription: ");
25             __isoc99_scanf("%127s",pbVar1->description);
26             printf("input book price (won): ");
27             __isoc99_scanf("%lu%c",&pbVar1->price);
28             books[local_c] = pbVar1;
```



 Rename Local Variable ✕

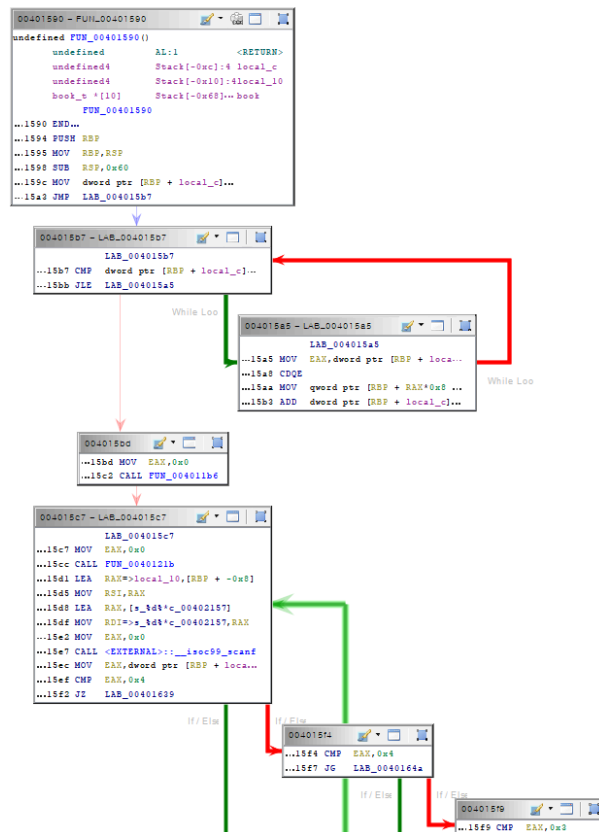
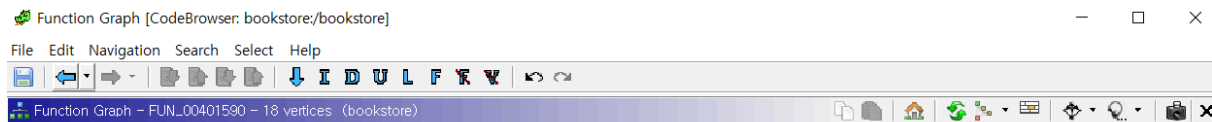
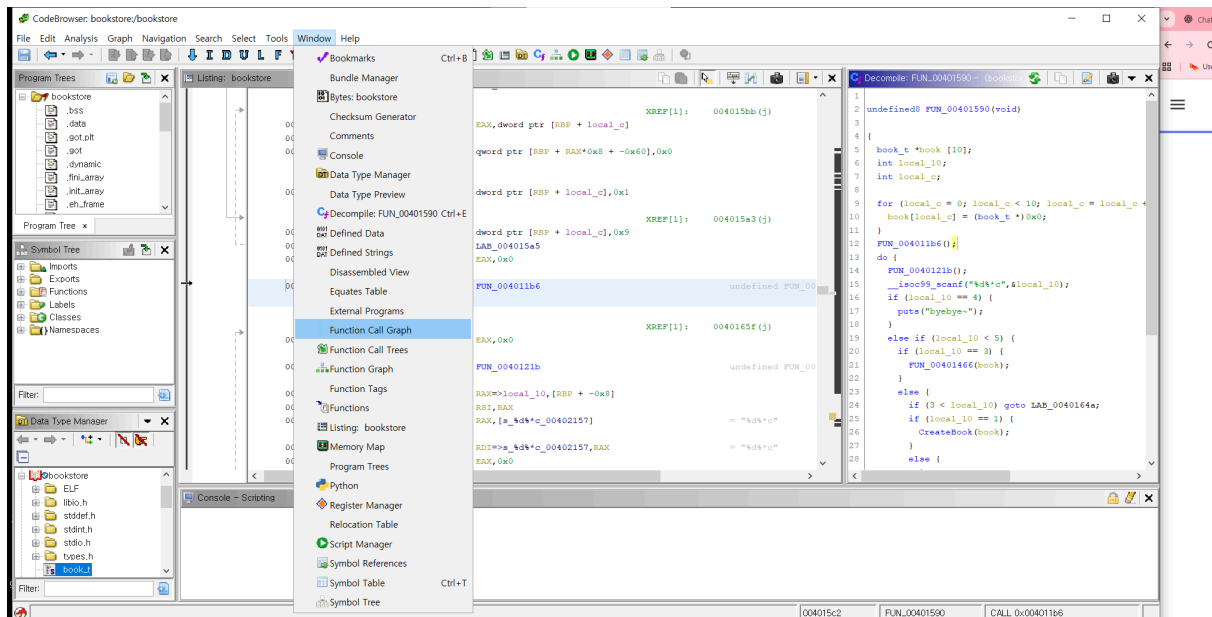
Rename local_68:

 Rename Local Variable ✕

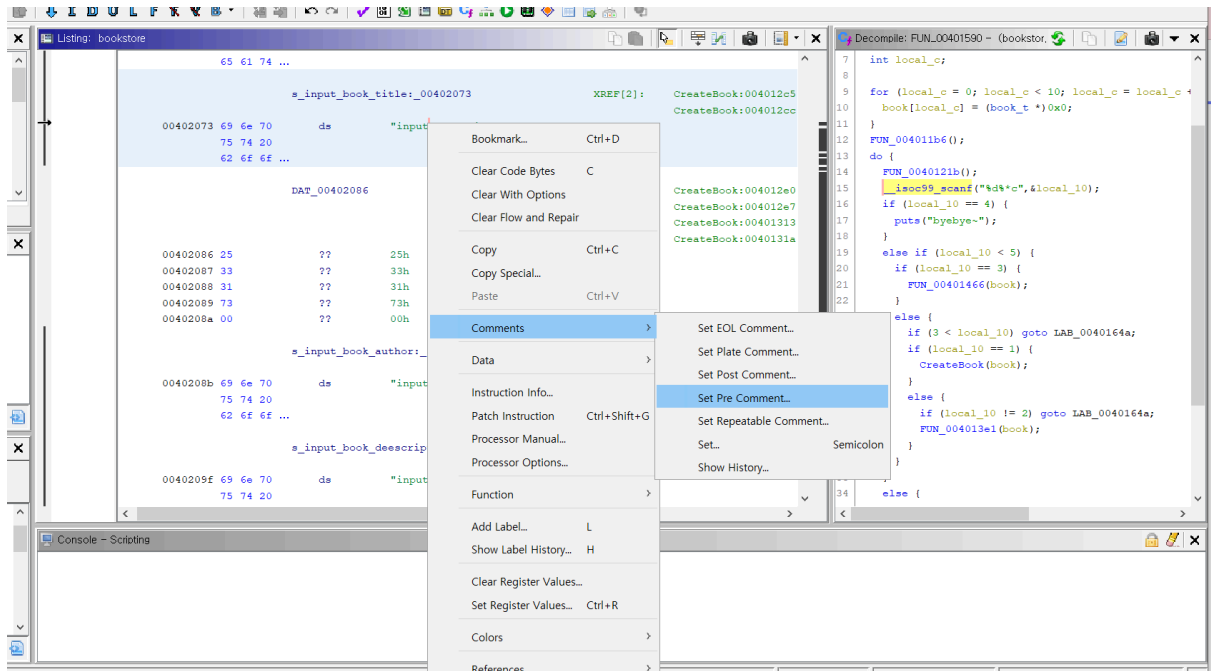
Rename local_68:

```
Decompile: FUN_00401590 - (bookstor,      
```

```
1
2 undefined8 FUN_00401590(void)
3
4 {
5     book_t *book [11];
6     int local_10;
7     int local_c;
8
9     for (local_c = 0; local_c < 10; local_c = local_c + 1)
10         book[local_c] = (book_t *)0x0;
11 }
12 FUN_004011b6();
13 do {
14     FUN_0040121b();
15     __isoc99_scanf("%d%c",&local_10);
16     if (local_10 == 4) {
17         puts("byebye~");
18     }
19     else if (local_10 < 5) {
20         if (local_10 == 3) {
21             FUN_00401466(book);
22         }
23         else {
24             if (3 < local_10) goto LAB_0040164a;
25             if (local_10 == 1) {
26                 CreateBook(book);
27             }
28             else {
```

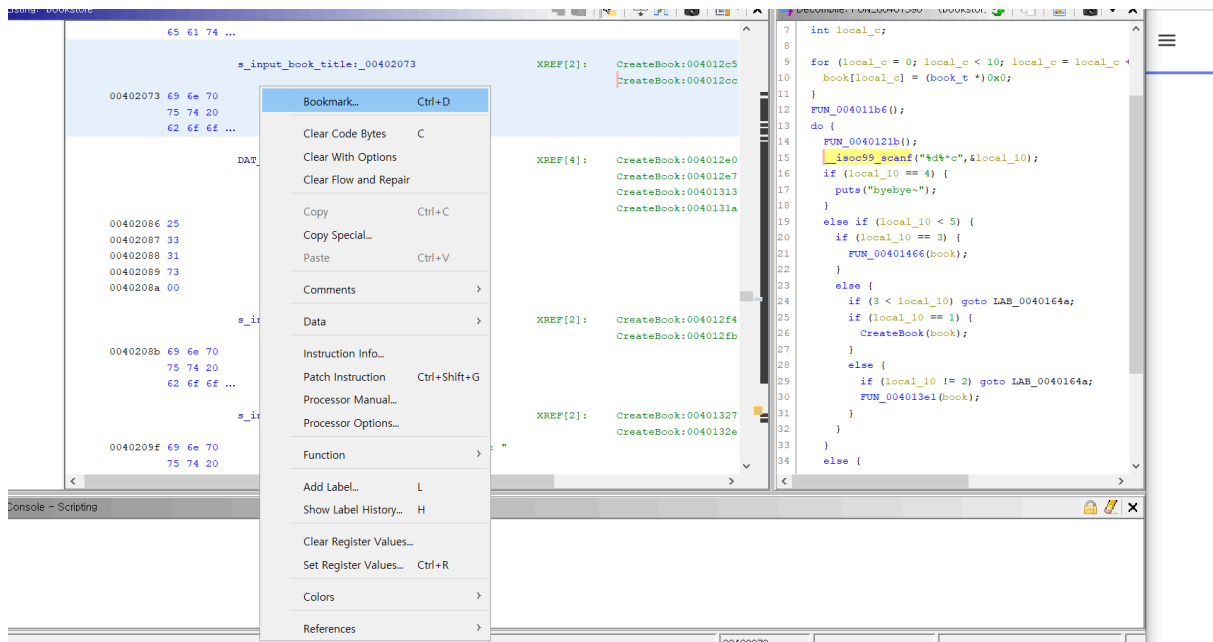


window > graph



주석 기능

- EOL Comment
 - 맨 오른쪽에 주석 표시
- Plate Comment
 - ***** 등 블록으로 표시되는 주석
- Post Comment
 - 명령어 위에 표시되는 주석



그러면 Figure41과 같은 창이 나옵니다. 북마크 카테고리를 설정하거나 설명을 적을 수 있습니다.

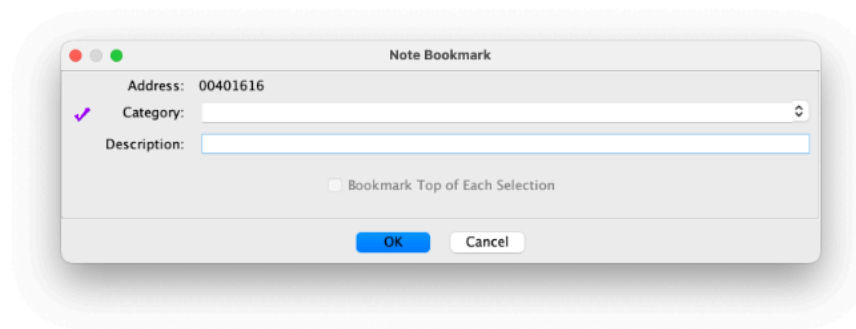
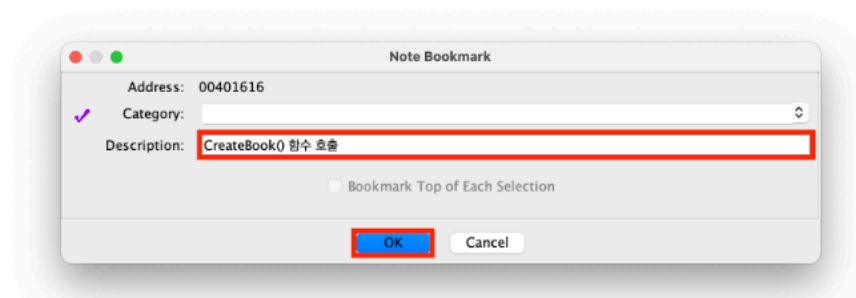


Figure 41. 북마크 추가 (2)

Figure42와 같이 설명을 적고 **OK**를 누릅니다.



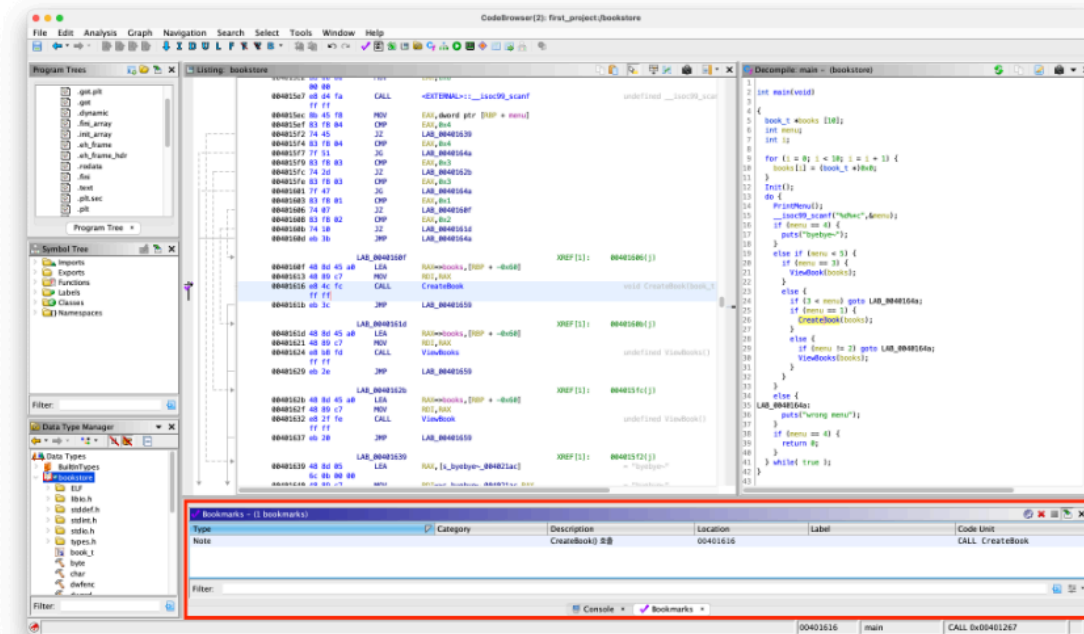
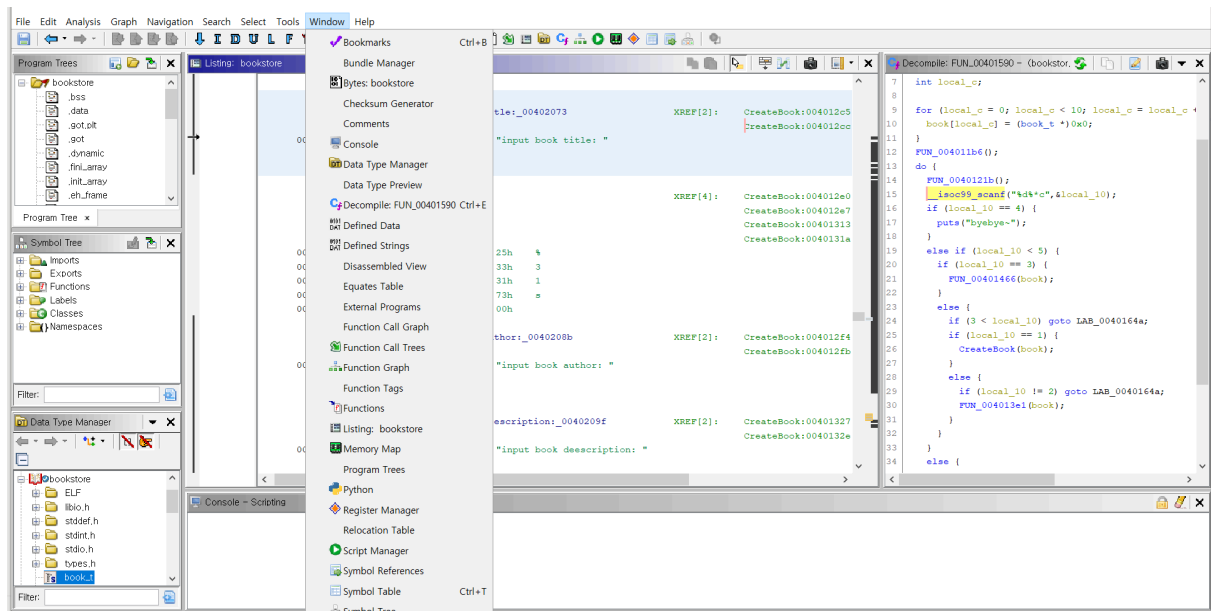
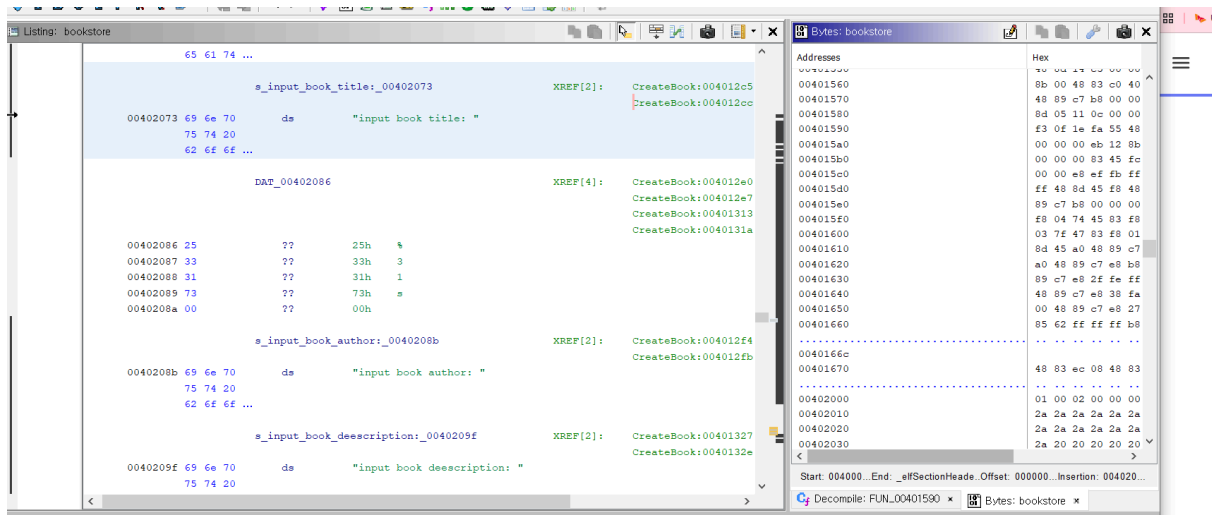


Figure 43. 북마크 추가 (4)

북마크 기능 통해 원하는 위치로 하단의 북마크 링크 클릭 통해 언제든지 복귀



Windows > Bytes: [프로그래밍 ex)bookstore] 클릭



오른쪽에 바이트 형식으로 확인 가능